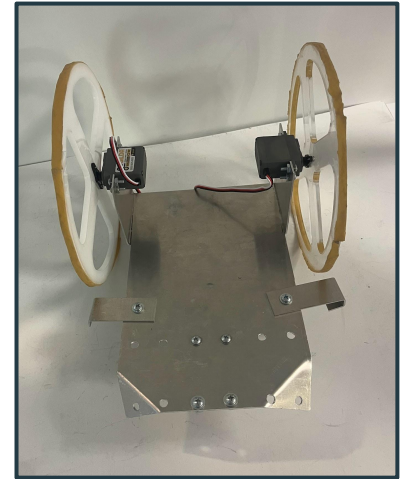
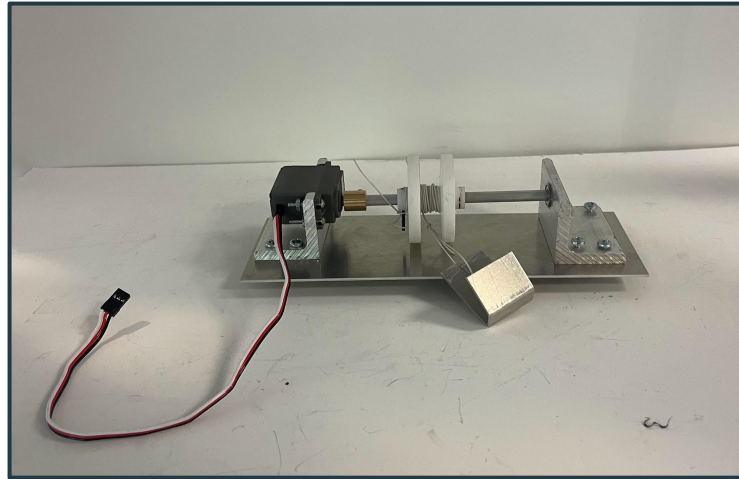


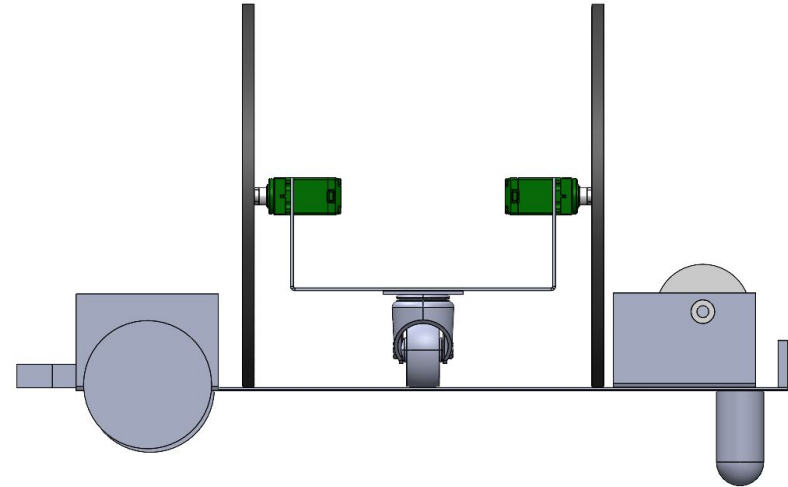
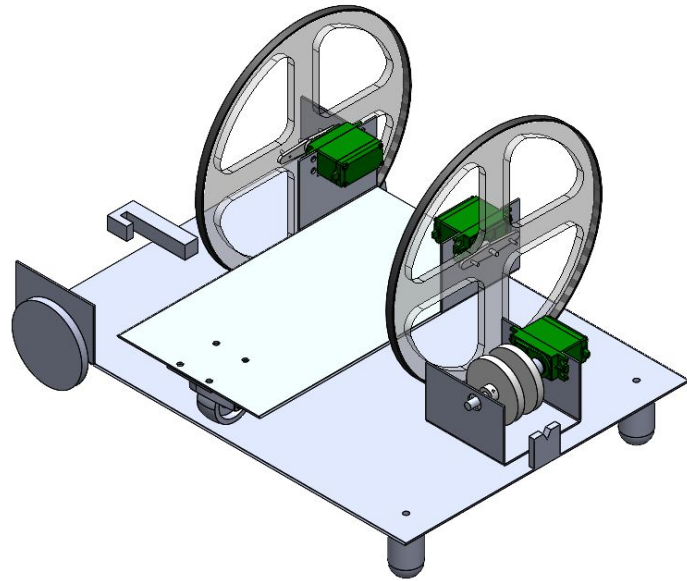
2.007

Oral Design Review

Jaden McKee
04/16/2026
Becker Thursday Lab 2-5



The Vision



Why this Vision?

Speed Bot (400 Pts)

- + Tire pull (2x)
- + Champagne Swing (1.3x)
- = Total Point Goal: 1040

Few Failure Modes
Lots of opportunity to
pivot

Design Requirements

- Complete a Lap in less than 25 seconds
- I must be able to complete multiple laps without crashing
- Winch must be able pull the tire in < 60 Seconds
- Speed Bot must fit in a 12" x 8" x 9" box (Track Legal + Fit on Winchbot)
- Winch Bot must fit in a 16" x 12" x 3" box (Robot Legal)

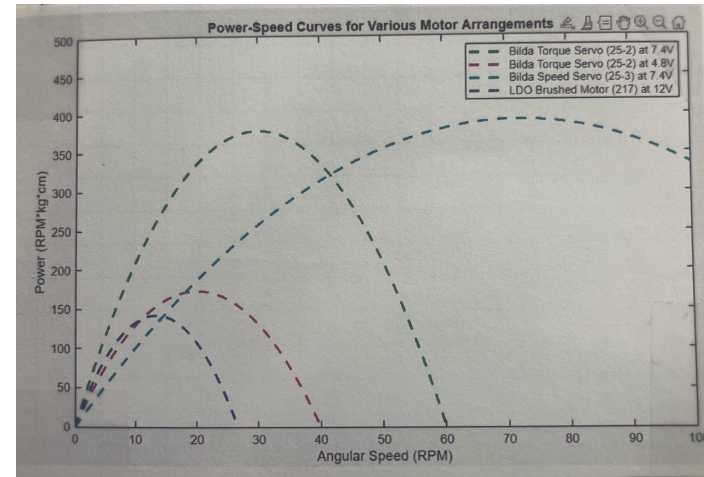
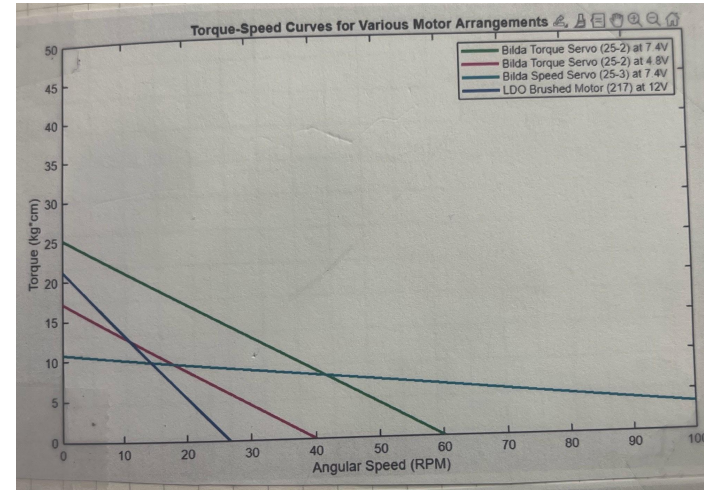
Speed Bot

- 8" Wheels, quarter inch acrylic with rubber strips superglued to outside
- 2 Speed Servos
- Caster Wheel in Front
- Crash Guards in front of wheels



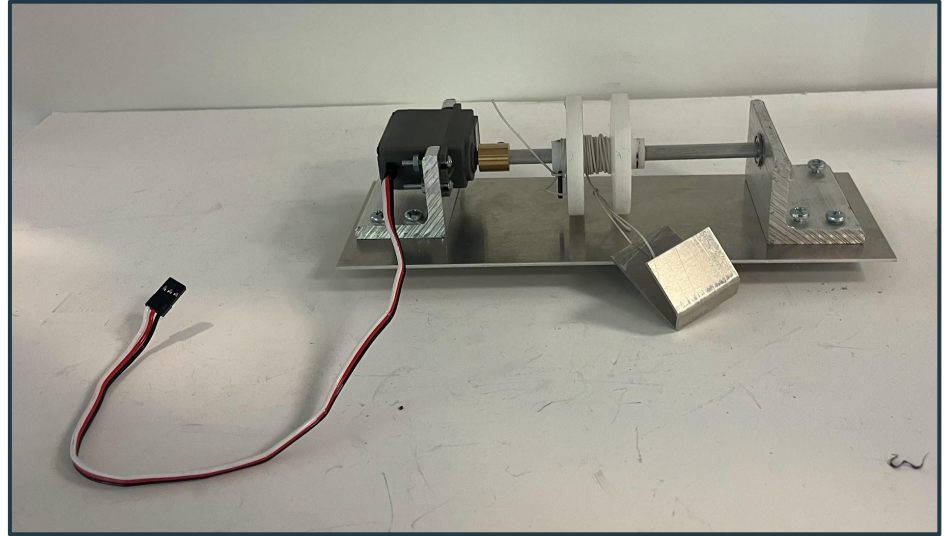
Justification

- Equivalent Gear Ratio = ~ 3
- Better Power Transmission
- Simpler
- Crash Guards Make Driving Easier
- Weight Concentrated in back to prevent slipping + Less torque on gradient slope



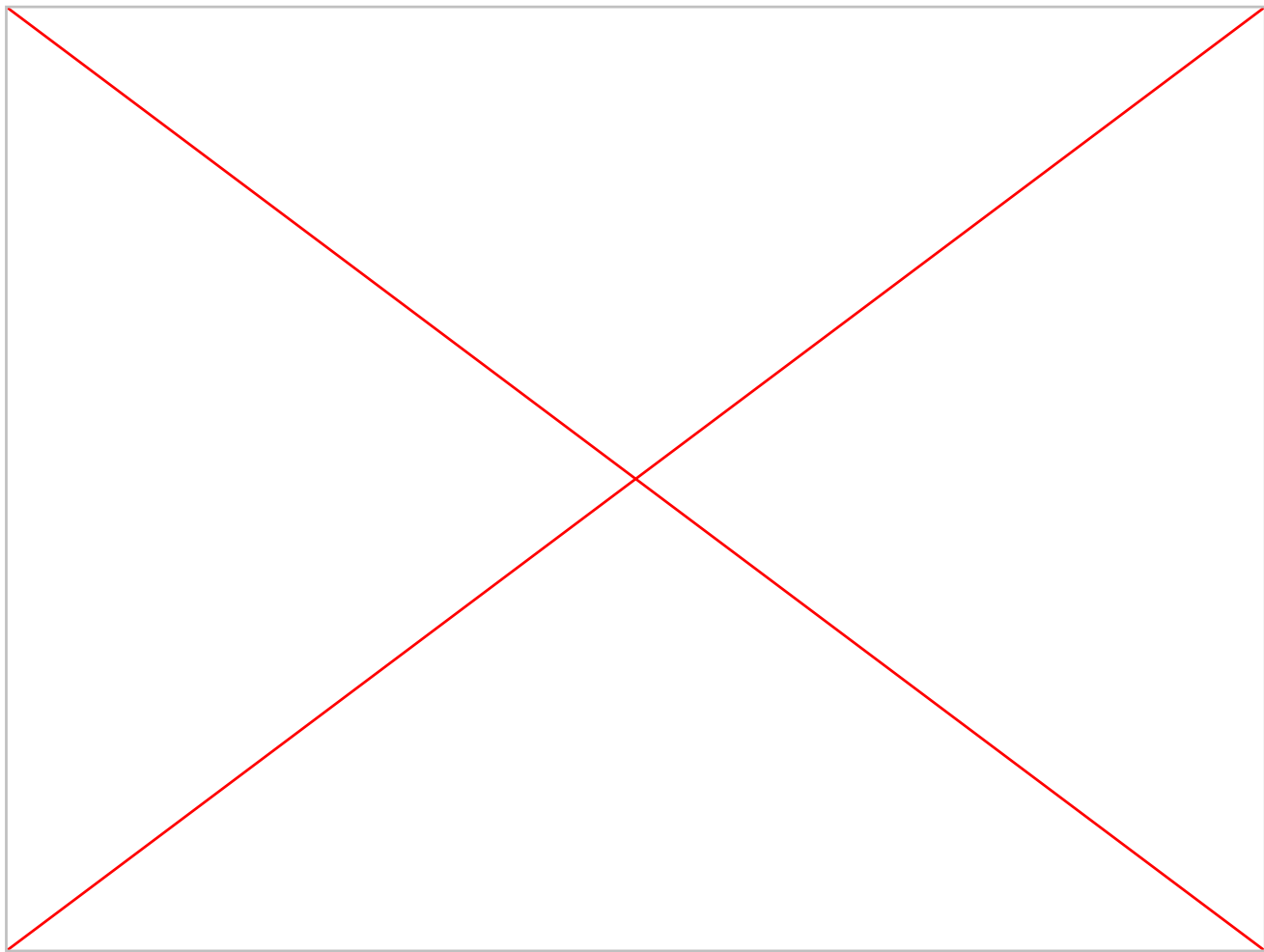
Winch

- 1x Torque Servo
- 1x Pulley, Drum Radius 0.25"
- 1x Ball Bearing
- 2x Brackets
- 2x Hooks



Justification

- Can pull over 2000N, Only needs ~100N to pull tire
 - Spring Scale Test, Motor Specs
- Sufficient Torque should not be a problem
- Back driving may be a concern, if so, I can ratchet my pulley



What is Left

		Design + Manufacture Winch	Design + Manufacture Ratchet for Winch	Build Chasis + Drive train for Winch	Milestone 8 Powerpoint	Champagne Tilt	Manufacture Final Line Follower Chasis	Code Line Follower + Test on old Robot	Milestone 9 Checklist	Manufacture Extra Wheels			
Lab	4/1												
	4/2	~15 hours	ooo										
	4/3												
	4/4												
	4/5												
	4/6												
	4/7		~5 hours										
	4/8												
Lab	4/9	xxx		~4 hours									
	4/10	xxx		xxx									
	4/11	xxx	xxx	xxx									
	4/12	xxx	xxx	xxx									
	4/13	xxx	xxx	xxx									
	4/14		xxx	xxx	~2 hours								
	4/15												
	4/16					~12 hours							
Lab (Milestone 8)	4/17												
	4/18												
	4/19												
	4/20												
	4/21												
	4/22						~5 hours	~18 hours					
	4/23												
	4/24												
Lab	4/25												
	4/26												
	4/27												
	4/28								~ 2 hours				
	4/29									~ 2 hours			
	4/30												
Lab													

KEY:

Planned Not Started

Planned in Progress

Planned Complete

Not Started

In Progress

Complete

xxx Behind Schedule

ooo Ahead Schedule

Questions?